

Purple Team Outline

Week 05: Network Detection & Defense Engineering

Objective: To design, implement, and test custom detection signatures for modern network threats.

Task 01: IDS vs IPS Logic

- **Description:** Determine where to place defensive appliances and how to configure their response actions.
- **Technical Requirements:**
 - Define the placement strategy for an IDS (Out-of-band) vs. an IPS (In-line).
 - Deploy Suricata or Zeek in the lab environment cross-segment traffic.
- **Learning Goal:** Understand the operational impact of "Block" (IPS) vs. "Alert" (IDS) modes.
- **What to Document :**
 - **Day 1:**
 - Install **Suricata** on Kali
 - Definition of IDS & IPS.
 - Difference table
 - Screenshot of installation
 - Output of: **ip a**
 - Run **Suricata** & Generate normal traffic.
 - Command used
 - Screenshot of Suricata running
 - Log file explanation: **/var/log/suricata/fast.log**
 - Explain: *Why IDS does not block traffic ?*
 - **Day 2:**
 - Install hping3 & perform **SYN** flood.
 - Attack Command

- Alert logs (proof)
- Explanation: What is SYN Flood? Why IDS only alerts?
- Enable **IPS** mode and understand traffic blocking.
- Configure iptables & Run Suricata in IPS mode.
- Commands used.
- Screenshot of IPS running.
- Explain: What is in-line placement? How IPS blocks traffic.
- **Day 3:**
 - Compare behavior of IDS vs IPS.
 - Run SYN flood again in: IDS mode, IPS mode.
 - Table comparison: Blocking & Detection.
 - Log comparison.
 - Observation: Did traffic stop in IPS?
 - Summarize learning and real-world understanding.
 - Analyze results
 - Answer conceptual questions
 - What happens in IDS mode?
 - What happens in IPS mode?
 - Which is better and why?
 - Real-world use cases: Where IDS is preferred & Where IPS is required ?

Task 02: Custom Detection Rule Creation

- **Description:** Transition from using default signatures to writing custom defense logic.
- **Technical Requirements:**
 - Create practical detection rules for previously practiced attacks (e.g., SYN Floods or ARP Spoofing).
 - Verify the rules by re-running the attacks and confirming alerts in the management console
- **Learning Goal:**
 - Learn the syntax of defense-converting an attacker's TTP (Tactics, Techniques, and Procedures) into a working signature.
- **What to Document :**
 - **Day 4:**
 - Understanding Rule Structure & Learn how detection rules work in Suricata
 - Open rules file: **sudo nano /etc/suricata/rules/local.rules**
 - Study basic rule format: **action protocol source → destination (options)**
 - Explain rule components: **alert, tcp/icmp, any any, msg, sid, rev.**
 - Screenshot of rules file
 - Write 2 example rules (theoretical)
 - First Custom Rule (ICMP Detection), Create and test a simple rule.
 - Add rule → Run Suricata → Generate Traffic.
 - Rule used
 - Command output
 - Log evidence: **/var/log/suricata/fast.log**
 - Explain: How the rule triggered & What ICMP is ?
 - **Day 5:**
 - TCP-Based Rule (Port Detection), Detects specific TCP traffic.
 - Add rule: **alert tcp any any -> any 80 (msg:"HTTP Traffic Detected"; sid:1000003; rev:1;)**

- Rule explanation: ***curl http://example.com***
- Screenshot of alert logs
- Explain: What is port 80? Why HTTP is important.
- Attack Detection Rule (SYN Flood), Convert attack into detection logic.
- Add rule: ***alert tcp any any -> any 80 (msg:"SYN Flood Detected"; flags:S; threshold:type both, track by_src, count 20, seconds 1; sid:1000004; rev:1;)***
- Install tool: ***sudo apt install hping3***
- Run attack: ***sudo hping3 -S -p 80 --flood <target-ip>***
- Rule breakdown (flags, threshold)
- Attack command
- Alert logs (proof)
- Explain: What is SYN Flood? How rule detects it
- **Day 6:**
 - ARP Spoofing Detection, Detecting network-level attacks.
 - Add rule: ***alert arp any any -> any any (msg:"ARP Spoofing Detected"; sid:1000005; rev:1;)***
 - Install tool: ***sudo apt install dsniff***
 - Run: ***sudo arpspoof -i eth0 -t <victim-ip> <gateway-ip>***
 - Rule used
 - Attack explanation
 - Logs showing detection
 - Explain: What is ARP Spoofing? Why it is dangerous
 - IPS Conversion (Block Instead of Alert), Convert detection into prevention.
 - Change rule: ***drop tcp any any -> any 80 (msg:"Blocked SYN Flood"; flags:S; sid:1000006; rev:1;)***
 - Run Suricata in IPS mode: ***sudo suricata -c /etc/suricata/suricata.yaml -q 0***
 - Re-run SYN flood.
 - Difference between **alert** and **drop**
 - Proof that traffic is blocked
 - Logs comparison